



Homework 6

Deadline: 23 November 2025, 23:30.

All solutions must be in a single PDF file and uploaded to the LMS portal.

Consider an Asian call option with payoff at time T and strike K . Assume the underlying asset S_t follows a geometric Brownian motion in risk-neutral measure:

$$dS_t = \sigma(t)S_t d\widetilde{W}_t.$$

where $\sigma(t)$ — deterministic function. Consider the quantity $M_t = \frac{1}{t} \int_0^t S_u du$ and

$$U_t = \frac{tM_t + (T - t)S_t}{T}.$$

1. (0.2 point) Is it true that $U_0 = S_0$ and $U_T = M_T$? Justify your answer.
2. (0.2 point) Can U_t be replicated using a position in asset S over an infinitesimal interval $[t, t + dt]$? If yes, what position is required for replication?
3. (0.3 point) Find the stochastic differential equation for U_t .
4. (0.3 point) Assuming that $S_t/U_t = 1$, find the price of the Asian call option. Compare the obtained price with the price of a vanilla European call option with the same parameters.